Vigneshkumar Venugopal

Tempe, AZ (Open to nationwide relocation)
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Education

Arizona State University, Tempe, AZ (MS, Data Science) (CGPA: 3.56/4.00) Aug 2025 – May 2027
Vellore Institute of Technology, Vellore, India (Btech, Computer Science) Aug 2020 – May 2024

Skills

Machine Learning & AI: PyTorch, TensorFlow, MLX, Transformers, LLMs, Deep Neural Networks, Multi-Agent Systems, Computer Vision (U-Net, CNNs), Sequence Modelling (LSTM, RNNs), XGBoost, LightGBM, Scikit-learn, SHAP, Optuna, Matlab, Julia, MLOps, RAG, Exploratory data analytics, statistical modeling, data visualization, data collection, Parameter-Efficient Fine-Tuning (PEFT), LoRA, QLoRA, 4-bit Quantization (NF4), Double Quantization, Instruction Tuning, System Prompt Engineering, GGUF Model Formats, GPTQ
Tools & Frameworks: Docker, Kubernetes, Git, GitHub Actions (CI/CD), Pytest, FastAPI, TensorBoard, Google Agent Development Kit, HuggingFace, tiktoken, NumPy, Pandas, Keras, Scikit-learn, Matplotlib, Seaborn, OpenAI API, Hadoop Map/Reduce, Spark, Cassandra, MlFlow, CUDA, CuDNN, CuBLAS, Bash/Shell Scripting, Ollama, Nemo Agent Toolkit, llama.cpp, Gradio, Weights & Biases (W&B), OpenCV, PIL, TensorRT, Triton Inference Server, TensorFlow Lite, ONNX, ExecuTorch, FSDP, vLLM, RAGAS, RLHF, DPO, MLX-LM
Languages: Python, Java, C++, C, Bash/Shell Scripting
Data & Storage: SQL, MongoDB, Vector databases (Elasticsearch, FAISS, Chroma, Milvus, Pinecone), CockroachDB, Kafka, PostgreSQL, MySQL, Redis, Celery
Cloud: GCP, VertexAI
Operating systems: Linux, Windows

Projects

NeuralScholar — 3B Language model finetuned to research and explain ML/DL concepts (github.com/vignesh-kumar-v/NeuralScholar)

- **Engineered a Specialized 3B ML Research LLM:** Fine-tuned **Qwen2.5-3B** using **QLoRA (4-bit NF4 quantization)** with custom system prompts and inference pipelines, achieving an **18% improvement in domain-specific technical accuracy** while cutting GPU memory overhead by **65% (<6 GB VRAM)** to enable edge deployment.
- **Architected an Automated Multi-Source Data Pipeline:** Built an end-to-end ETL system ingesting **100K+ samples** from ArXiv, StackExchange, and Distill.pub; implemented **MinHash LSH deduplication** and pattern-based code filtering, removing **~35% redundant data** and accelerating training convergence
- **Optimized for Cross-Model Synergy & Deployment:** Engineered a two-stage routing workflow pairing local SLMs with frontier LLMs (Gemini/Claude) and exported quantized **GGUF weights via llama.cpp/Ollama**, achieving **~45 tokens/sec local inference** and reducing external API token costs by **~60%**

NanoLLM – Custom Transformer Implementation (github.com/vignesh-kumar-v/LLMs)

- **End-to-End LLM Development:** Engineered a decoder-only **GPT-2 style transformer from scratch in PyTorch** and trained on TinyStories; built custom data pipelines with sliding-window batching and **mixed-precision (BF16)** training loops, achieving **>2.5× faster training throughput** and stable convergence across **10M+ tokens**.
- **Custom CUDA Kernel Optimization:** Authored and optimized 3 progressive versions of fused LayerNorm CUDA kernels in C++/CUDA using **shared-memory tree reduction, Welford's algorithm with warp shuffles, and vectorized float4 memory access**, achieving a **1.7× speedup** and **~40% memory bandwidth efficiency gain** over native PyTorch baselines.
- **Profiling & Performance Telemetry:** Designed an automated benchmarking suite using **NVIDIA Nsight Compute (ncu)** with **NVTX markers** to profile register pressure and kernel latencies; built telemetry for gradient/weight norms, achieving **sub-millisecond profiling overhead** and optimizing peak VRAM footprint by **~30%**.

AEGIS-multi-Agent-system (github.com/vignesh-kumar-v/AEGIS-Multi-Agent-system)

- **Designed** a distributed multi-agent system using **Google's Agent Development Kit** to automate real-time flood monitoring, alert triage, and emergency intelligence synthesis
- **Architected** a modular "Agent Mesh" with specialized roles (FloodMonitor, TriageAgent, Medical/Infra Sentinels) coordinated via **Python's asyncio** to enable non-blocking, parallel situational analysis
- **Engineered** a production-grade workflow including separate modules for agent lifecycles, external tool integration, and consolidated mission reporting, open-sourced to demonstrate scalable agentic system design

TelcoFlow — Customer Churn Prediction Pipeline (github.com/vignesh-kumar-v/TelcoFlow-CI-CD)

- **Designed an end-to-end MLOps pipeline for customer churn prediction**, utilizing **Optuna** for Bayesian hyperparameter optimization across **LightGBM and XGBoost** to achieve an **ROC-AUC of 0.86 (12% lift over baseline)**, integrating **SHAP** to quantify key feature attributions for business stakeholders

- **Engineered a production-grade ML inference system** with **FastAPI** and **Docker**, delivering sub-**30ms API response latency** and establishing a fully automated **CI/CD pipeline via GitHub Actions** that executes schema validation, unit tests, model retraining, and artifact versioning on every commit
- **Built a modular, reproducible data engineering pipeline** processing **7,000+ customer records**, implementing automated cleaning, stratified train/validation/test splits, and **statistical drift detection (Evidently AI/KS-tests)** to alert on covariate shift and maintain high reliability in production

VulnTriage-LLM — An automated CVE triage framework leveraging fine-tuned SecBERT, LLMs, and multi-tool agentic pipelines to predict CVSS v3.1 severity from natural language descriptions (github.com/vignesh-kumar-v/VulnTriage-LLM)

- **Architected a Multi-Model Triage Engine:** Engineered a comparative vulnerability classification system using PyTorch and Hugging Face Transformers to evaluate zero-shot LLM inference against a specialized, fine-tuned cybersecurity encoder (**SecBERT**) for mapping text descriptions to CVSS v3.1 severity levels.
- **Designed End-to-End Data & Fine-Tuning Pipelines:** Built robust data engineering components for exploratory data analysis (EDA) and custom preprocessing; implemented sequence classification fine-tuning routines, logging structured metrics to JSON, and establishing a baseline Macro F1 score of **0.4253**.
- **Developed an Agentic Automation Layer:** Implemented a multi-tool agentic workflow that consumes raw model outputs and autonomously generates structured, analyst-ready triage reports to significantly minimize manual patch-prioritization overhead for security operations.

PaperCut — An autonomous multi-agent orchestration system powered by LangGraph, Vertex AI, and NVIDIA NIM to translate ArXiv ML papers into validated PyTorch scaffolds and optimized CUDA blueprints (github.com/vignesh-kumar-v/PaperCut)

- **Architected Stateful Multi-Agent Pipelines**: Designed and deployed a cascading, stateful agent network using **LangGraph** and **FastAPI** to orchestrate four specialized frontier LLMs (Gemini 2.5 Flash/Pro via Vertex AI, Qwen2.5-Coder, and Nemotron Super), implementing automated scope-validation layers parse research papers, achieving **>95% validation accuracy** across **50+ arXiv architectures**
- **Automated Code Gen & CUDA Optimization Engineering:** Engineered parallel generation pipelines that utilize **Qwen2.5-Coder 32B** to produce structured **PyTorch** scaffolds (`model.py`, `train.py`, `dataset.py`) based on an extracted JSON R*esearchContract*, coupled with **Nemotron Super 49B** via NVIDIA NIM to output annotated CUDA C++ stubs, reducing manual boilerplate generation time by **~75%** with zero structural syntax errors
- **Built Real-Time Streaming & Cloud Infrastructure:** Constructed a fully non-blocking backend leveraging Server-Sent Events (SSE) to deliver sub-**200ms TTFT (Time-To-First Token)** and support **10+ concurrent agent streams** via GCS

Engram — Hierarchical Memory System for LLM Coding Agents (github.com/vignesh-kumar-v/Engram)

- Designed and built a **biologically-inspired hierarchical memory architecture** for LLM coding agents — **episodic buffering**, **LLM-driven consolidation** (promote/compress/discard reasoning), and **long-term semantic storage** — to solve the cross-session memory loss and contradiction-blindness common in existing agent memory systems like **MemGPT**.
- Implemented the full pipeline in **Python** (**LangGraph**, **SQLite**, **Qdrant**, **Ollama/Qwen2.5-7B**) including a background **consolidation agent** with **contradiction detection** and persistence, backed by **336 passing unit/integration tests** across the core system and benchmark suite.
- Designed and ran **LHMBench**, an original **6-scenario benchmark** comparing the system against naive RAG and stateless baselines; achieved a **perfect 100% score** with the only system to detect memory contradictions (**1.0 vs. 0.0** for both baselines), and validated robustness via a custom **multi-session evaluation harness** across **8 simulated project lifecycles**.

FlashContext — Full-Stack RAG System for Academic Research (github.com/vignesh-kumar-v/FlashContext)

- **Architected an end-to-end automated RAG ingestion pipeline** that uses a local LLM (Ollama) for query keyword extraction, queries ArXiv via **dual-strategy ranking** (relevance + recency), and automatically vectorizes ~25 papers per session into Qdrant in **<15 seconds**, improving query-document relevance by **~30%** over standard keyword search
- **Engineered a custom layout-aware PDF chunking algorithm** using PyMuPDF to calculate relative font-size baselines and split text along semantic section boundaries, incorporating a **LangChain RecursiveCharacterTextSplitter** fallback and evaluating performance with **RAGAS/TruLens** to achieve a **22% increase in retrieval precision (Hit Rate@5)** over fixed-size chunking
- **Designed a real-time, multi-tenant streaming architecture** combining FastAPI WebSockets, Celery async queues, and Redis Pub/Sub to deliver **>40 tokens/sec streaming throughput** with background re-indexing latencies under **2 seconds**

MLX-OCR-Unlimited — Native MLX Implementation of the Unlimited-OCR Model for Apple Silicon (github.com/vignesh-kumar-v/mlx-ocr-unlimited)

- **Engineered a High-Performance Native MLX Port:** Implemented a pure MLX inference pipeline for the 3.3B-parameter *Unlimited-OCR (DeepSeek-OCR)* model on Apple Silicon, eliminating PyTorch dependencies at runtime and achieving decode speeds up to ~175 tokens/sec (~6 GB memory overhead).
- **Optimized Vision & MoE Architecture Components:** Re-architected custom MLX components, including a stacked MoE layer using batched `gather_mm`, a specialized sliding KV cache with pinned image-prompt context, and bit-exact position embedding resizing matching PyTorch's bicubic-antialias interpolation via cached separable matrices.

- **Ensured Numerical Parity & Robust Validation:** Quantified output fidelity against the baseline CUDA implementation across complex 15-page document parsing tasks, validating high accuracy with a 0.9998 cosine output similarity in FP32 (0.99 in FP16) and introducing comprehensive MoE gating unit tests.

Work Experience

Education and Research Network of India (Chennai, India) May 2024 – June 2025

Internship

- **Architected** a multi-class Classification and Segmentation pipeline to identify glioma, meningioma, and pituitary tumors, achieving **96.4% classification accuracy** on real-world MRI datasets
- **Developed** a ResUNet semantic segmentation model achieving a **Dice Coefficient of 0.89,** automating tumor boundary delineation and reducing manual volume estimation time for radiologists by **70%**
- **Optimized** high-resolution NIfTI/DICOM data-loading pipelines using **MONAI**, resulting in a **35% reduction in training latency** and enabling efficient multi-GPU training

Research

A novel ensemble machine learning approach for optimizing sustainability and green hydrogen production in hybrid renewable-based organic Rankine cycle-operated proton exchange membrane electrolyser system (doi.org/10.1016/j.renene.2025.122369)

- **Designed and implemented a novel ensembled Transformer-LSTM deep learning architecture** to model the nonlinear, time-series dynamics of a complex hybrid renewable energy system. **Integrated multi-headed attention mechanisms with LSTM layers** to successfully capture both long and short-term temporal dependencies, **overcoming the vanishing gradient problems and hyperparameter sensitivity inherent in traditional Recurrent Neural Networks (RNNs)**
- **Engineered a robust data pipeline and conducted deep learning interpretability analysis using Python**. Prepared and preprocessed extensive simulation datasets utilizing **5-fold cross-validation for high generalizability**, and applied **integrated gradients analysis to determine feature importance**, ensuring the neural network's attributions logically aligned with underlying real-world constraints
- **Benchmarked model performance against baseline neural networks (FFNN, RNN, and standard LSTM) utilizing rigorous evaluation metrics (MSE, MAE, MAPE, and SMAPE)**. The proposed **Transformer-LSTM ensemble achieved state-of-the-art predictive accuracy with an overall Mean Absolute Percentage Error (MAPE) of 0.74%**, **significantly outperforming the standard LSTM (0.89%), RNN (1.69%), and FFNN (2.1%) in multi-variable time-series forecasting**

Analysis and multi-objective evolutionary optimization of Solar-Biogas hybrid system operated cascade Kalina organic Rankine cycle for sustainable cooling and green hydrogen production (doi.org/10.1016/j.enconman.2023.117999)

- **Developed** and **simulated** complex computational models in **Python**, integrating the **CoolProp library** to automate dynamic **thermodynamic property acquisition**. Engineered a robust programmatic pipeline to continuously evaluate the **energetic**, **exergetic**, **economic**, and **environmental (4E)** mathematical equations of a novel **solar-biogas hybrid system** under highly variable boundary conditions.
- **Architected** a **multi-objective evolutionary optimization framework** utilizing the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)** to resolve conflicting, non-linear system objectives. Successfully **optimized** highly constrained system variables across **400 generations** to maximize **energy utilization** and **exergy efficiency** while simultaneously minimizing total system cost, generating a comprehensive **Pareto frontier**.
- **Designed** and **implemented** a sophisticated **two-stage Multi-Criteria Decision-Making (MCDM) algorithmic pipeline** to systematically evaluate high-dimensional **Pareto optimization data**. Programmed the **CRITIC statistical method** for objective weight calculation, and deployed **TOPSIS** and **MABAC algorithms** for advanced data normalization, distance-based ranking, and the precise selection of the ultimate **optimal system configuration**.